1. Introduction

Wind loads are one of the major factors affecting the safety and stability of multistorey buildings. Traditional design methods generally use the provisions of IS 875 (Part 3) to estimate wind loads. However, actual wind behavior around buildings is complex and depends on factors such as building geometry, surrounding structures, and local environmental conditions.

This project focuses on evaluating the wind-induced performance of a G+9 residential building using Computational Fluid Dynamics (CFD) and structural analysis. ANSYS Fluent is used to simulate wind flow around the building, while STAAD.Pro is used to assess the structural response under wind loading conditions.

2. Problem Definition

2.1. Problem Statement:

Conventional wind load calculations based on IS 875 use generalized pressure coefficients that may not accurately represent actual wind pressure acting on a specific building. This can lead to inaccurate estimation of structural behavior. Therefore, there is a need to evaluate realistic wind pressures using CFD analysis and assess their impact on building performance.

2.2. Background Information: (literature review)

Several researchers have studied wind effects on buildings.

Ali and Moon (2018) discussed structural systems used in tall buildings and emphasized the importance of resisting lateral loads.

Wijesooriya et al. (2023) reviewed CFD applications in wind engineering and highlighted the importance of turbulence modeling and mesh quality.

Hu et al. (2024) studied wind effects on low-rise buildings and found that pressure concentrations are generally higher near corners and edges.

Most previous studies focused either on CFD analysis or structural analysis individually. Limited work has combined CFD-based wind pressure analysis with structural performance evaluation and retrofit design for mid-rise residential buildings.

3. Objectives

3.1. Primary Objectives:

1. Develop a 3D model of a G+9 residential building.
2. Perform CFD analysis using ANSYS Fluent.
3. Determine wind pressure distribution and pressure coefficients
4. Compare CFD results with IS 875 values.
5. Evaluate structural performance under wind loading.

3.2. Secondary Objectives:

1. Identify critical structural zones.
2. Analyze displacement, storey drift, bending moments, and shear forces.
3. Propose suitable retrofit measures for performance improvement.

4. Methodology

4.1 Approach: The project follows an integrated CFD and structural analysis approach.

Flow: Building Selection → STAAD.Pro Modeling → Load Application → CFD Modeling in ANSYS Fluent → Wind Flow Analysis → Pressure Distribution Extraction → Structural Analysis → Identification of Critical Zones → Retrofit Design → Reanalysis

4.2 Procedures:

1. Developed a G+9 building model in STAAD.Pro.
2. Applied dead load, live load, and wind load.
3. Created the CFD model in ANSYS Fluent.
4. Generated computational mesh.
5. Applied wind speed of 47 m/s (Wind Zone 4)
6. Obtained pressure and velocity contours.
7. Extracted pressure coefficients.
8. Applied CFD pressures to the structural model.
9. Evaluated displacement, support reactions, bending moments, and shear forces.
10. Proposed retrofit measures and assessed performance.

5. Project Execution

5.1 Planning and Design: Initially, a suitable G+9 residential building configuration was selected. Structural dimensions and loading conditions were finalized according to IS 875 provisions. CFD domain dimensions and mesh requirements were also planned.

5.2 Implementation: The building was modeled in STAAD.Pro and analyzed under wind loading. CFD simulations were performed in ANSYS Fluent using a wind velocity of 47 m/s. Pressure contours and velocity contours were obtained and used for structural performance evaluation

6.Tools and Techniques Used

Tools:

1. ANSYS Fluent-CFD simulation and wind pressure analysis
2. STAAD.Pro-Structural modeling and analysis
3. IS 875 (Part 3)-Wind load calculations
4. AutoCAD-Geometry preparation.

Techniques:

- Computational Fluid Dynamics (CFD)
- Wind Pressure Coefficient Evaluation

- Structural Performance Assessment
- Comparative Analysis with IS 875

These techniques were selected to obtain realistic wind loading conditions and evaluate the structural response accurately.

7. Partial Results

7.1 Initial Findings:

1. Maximum pressure occurred on the windward face.
2. Negative pressure zones developed on the leeward face.
3. High pressure gradients were observed near building corners.
4. Wake regions and flow separation were observed behind the building
5. Alterative Improvements: Describe any iterations or improvements made based on initial results. Explain the rationale behind these changes.

8. Results and Discussion

CFD Results

1. Maximum positive pressure occurred on the windward face.
2. Negative pressure (suction) was observed on the leeward face.
3. Vortex formation and recirculation zones developed behind the building.
4. Higher velocity was observed along building edges.

Structural Results

- Parameter Structure 2 Structure 1
- Max X Displacement 7.29 mm 4.09 mm
- Max Resultant Displacement 16.83 mm 25.52 mm

Support Reactions

Parameter	Model 2	Model 1
Max Fx	26.94 kN	22.11 kN
Max Fy	4733.63 kN	2875.34 kN
Max Fz	51.29 kN	43.99 kN

9. Prototype Description

The developed prototype consists of:

1. A 3D structural model of a G+9 residential building in STAAD.Pro.
2. A CFD simulation model in ANSYS Fluent.

3. Wind pressure contours, velocity contours, and streamline plots.
4. Structural analysis results including displacement and reaction forces.
5. Development Process
6. The building geometry was first modeled in STAAD.Pro. A CFD domain was then created in ANSYS Fluent and meshed. Wind flow simulations were conducted, and the resulting pressure data was transferred to the structural model for analysis.

Testing and Validation

The CFD-derived pressure coefficients were compared with IS 875 pressure coefficients. The structural model was analyzed using these loads and validated by evaluating displacement patterns, support reactions, and deformation behavior.

10.Conclusion

10.1 Summary

This project evaluated the wind-induced behavior of a G+9 residential building using CFD and structural analysis. ANSYS Fluent was used to obtain realistic wind pressure distributions, while STAAD.Pro was used to assess structural response. The results showed that CFD analysis provides more accurate pressure predictions than conventional code-based methods. The study successfully identified critical structural zones and demonstrated the effectiveness of integrating CFD with structural analysis for safer and more reliable building design.

10.2 Personal Reflection

Through this project, I gained practical knowledge of wind engineering, CFD analysis, and structural modeling. I learned how ANSYS Fluent and STAAD.Pro can be integrated to evaluate the real behavior of buildings under wind loading. The project improved my understanding of structural safety, simulation techniques, and engineering decision-making, making it a valuable learning experience.